$$\frac{\partial L}{\partial \lambda} = 100 - u_1 - u_2 = 0$$

Starting with the first two equations, we see that $u_1 = u_2$ and λ drops out. From the third equation we can easily find that $u_1 = u_2 = 50$ and the constrained maximum value for u_3 is $u_3 = u_1 u_2 = 2500$

2) Set up the Lagrangian Equation

$$L = 4u_1^2 + 3u_1u_2 + 6u_2^2 + \lambda(56 - u_1 - u_2)$$

Take the first-order partials and set them to zero

$$\frac{\partial L}{\partial u_1} = 8u_1 + 3u_2 - \lambda = 0$$

$$\frac{\partial L}{\partial u_2} = 3u_1 + 12u_2 - \lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = 56 - u_1 - u_2 = 0$$

From the first two equations we get

$$8u_1 + 3u_2 = 3u_1 + 12u_2$$

$$u_1 = 1.8u_2$$

Substitute this result into the third equation

$$56 - 1.8u_2 - u_2 = 0$$

$$u_2 = 20$$

Thus $u_1 = 36$, $u_2 = 20$, $\lambda = 348$

Check Your Progress 2

1) For a given value of a, the function f are given by

$$\frac{\partial f}{\partial u} = -2u + 2a = 0 \Leftrightarrow u = a$$

The solution yields is a local (and global) maximum

Thus, we know that $u^*(a) = a$ and the value function at the optimum is

$$V(a) = f(u^*(a); a) = -a^2 + 2a^2 + 4a^2 = 5a^2$$

Hence, the derivative of the value function is given by

$$\frac{\partial V}{\partial a} = \frac{\partial f(u^*(a); a)}{\partial a} = \frac{\partial (5a^2)}{\partial a} = 10a$$

We could derive this directly using the envelope theorem

$$\frac{\partial V}{\partial a} = \frac{\partial f}{\partial a}\Big|_{u=u^*(a)} = 2u + 8a\Big|_{u=u^*(a)} = 2a + 8a = 10a\,.$$

2) The NDCQ condition is satisfied when $a \neq 0$, and the Lagrangian is given by

$$L = u + 3v - \gamma(u^2 + av^2 - 10)$$

By the envelope theorem, we have

$$\left(\frac{\partial f^*(a)}{\partial a}\right)_{a=1} = \left(-\gamma v^2\right)_{u=(1,3),\gamma=1/2} = -\frac{9}{2}$$

An estimate for $f^*(1.01)$ is therefore given by

$$f^*(1.01) \approx f^*(1) + 0.01 \cdot \left(\frac{\partial f^*(a)}{\partial a}\right)_{a=1} = 10 - 0.045 = 9.955$$

To find an exact expression for $f^*(a)$, we solve the first order conditions

$$\frac{\partial L}{\partial u} = 1 - \gamma \cdot 2u = 0 \Rightarrow u = \frac{1}{2\gamma}$$

$$\frac{\partial L}{\partial v} = 3 - \lambda \cdot 2av = 0 \Rightarrow v = \frac{3}{2a\gamma}$$

We substitute these values into the constraint $u^2 + av^2 = 10$, and get

$$\left(\frac{1}{2\gamma}\right)^2 + a\left(\frac{3}{2a\gamma}\right)^2 = 10 \Leftrightarrow \frac{a+9}{4a\gamma^2} = 10$$

This gives $\gamma = \pm\sqrt{\frac{a+9}{40a}}$ when $a > 0$ or $a < -9$. Substitution gives solutions for $u^*(\alpha), v^*(\alpha)$ and $f^*(\alpha)$. For $a = 1.01$, this gives $u^*(1.01) \approx 1.0045, v^*(1.01) = 2.9836$ and $f^*(1.01) \approx 9.9553.$

Check Your Progress 3

1) The function $f(u_1, u_2) = 3u_1^5 u_2 + 2u_1^2 u_2^4 - 3u_1^3 u_2^3$,

By definition of homogeneous function

$$f(tu_1, tu_2) = 3t^5 u_1^5 tu_2 + 2t^2 u_1^2 t^4 u_2^4 - 3t^3 u_1^3 t^3 u_2^3 = t^6 f(u_1, u_2)\,.$$

Hence, f is a homogeneous function of degree 6.

2) The function $f(u_1, u_2) = \alpha \ln(u_1) + \beta \ln(u_2) = \ln(u_1^\alpha u_2^\beta) = h(g(u_1, u_2))$,

where $g(u_1, u_2) = u_1^\alpha u_2^\beta$ and $h(u_1) = \ln(u_1)$.

Check Your Progress 4

1) The bordered Hessian of f is

$$=\begin{pmatrix} 0 & -u_2e^{-u_1} & e^{-u_1} \\ -u_2e^{-u_1} & u_2e^{-u_1} & -e^{-u_1} \\ e^{-u_1} & -e^{-u_1} & 0 \end{pmatrix}$$

$$D_1(u_1,u_2) = -u_2^2 e^{-2u_1} \le 0$$

We have $D_2(u_1,u_2) = u_2e^{-3u_1} \le 0$

and $D_2(u_1,u_2) = u_2e^{-3u_1} + e^{-u_1}(u_2e^{-2u_1} - u_2e^{-2u_1}) = u_2e^{-3u_1} \ge 0$

Both determinants are zero if $u_2 = 0$, so while the bordered Hessian is not inconsistent with the function's being quasi-concave, it does not establish that it is in fact quasi-concave either. However,

the test does show that the function is quasi-concave on the domain in which u_2 is restricted to be positive (rather than only non-negative).

2) $\dfrac{\partial f}{\partial u_1} = 6u_1^2, \dfrac{\partial f}{\partial u_2} = -12u_2$

$$\frac{\partial^2 f}{\partial u_1^{\,2}} = 12u_1, \frac{\partial^2 f}{\partial u_1 \partial u_2} = 0, \frac{\partial^2 f}{\partial u_1^{\,2}} = -12$$

$$H(u_1) = \begin{bmatrix} 12u_1 & 0 \\ 0 & -12 \end{bmatrix}$$

If $u_1 \le 0,\ 12u_1 \le 0$.

Determinant of $H(u_1) = -144u_1 \ge 0$.

Hence $H(u_1)$ is negative semi-definite for $u_1 \le 0$. Thus $f(u_1,u_2)$ is concave for $u_1 \le 0$.

19.1 EXERCISES

Q1. A firm products goods A and B. The price of A is 13, and the price of B is p. The profit function is $\pi(u,v) = 13u + pv - C(u,v)$, where

$$C(u,v) = 0.04u^2 - 0.01uv + 0.01v^2 + 4u + 2v + 500.$$

Determine the optimal value function $\pi^*(p)$. Verify the envelope theorem.

Ans. The profit function is $\pi(u,v) = 13u + pv - C(u,v)$, hence we compute

$$\pi(u,v) = -0.04u^2 + 0.01uv - 0.01v^2 + 9u + (p-2)v - 500$$

The first order conditions are

$$\pi_u = -0.08u + 0.01v + 9 = 0 \Rightarrow 8u - v = 900$$

$$\pi_v = 0.01u - 0.02v + p - 2 = 0 \Rightarrow u - 2v = 200 - 100p$$

This is a linear system with unique solution $u^* = \frac{1}{15}(1600 + 100p)$ and $v^* = \frac{1}{15}(-700 + 800p)$. The Hessian $\pi'' = \begin{pmatrix} -0.08 & 0.01 \\ 0.01 & -0.02 \end{pmatrix}$. is negative definite since $D_1 = -0.08 < 0$ and $D_2 = 0.0015 > 0$. We conclude that (u^*, v^*) is a (local and global) maximum for π.

Hence the optimal value function $\pi^*(p) = \pi(u^*, v^*)$ is given by

$$\pi\left(\frac{1}{15}(1600 + 100p), \frac{1}{15}(-700 + 800p)\right) = \frac{80p^2 - 140p + 80}{3}$$

and its derivative is therefore $\frac{\partial}{\partial p}\pi(u^*, v^*) = \frac{160p - 140}{3}$.

On the other hand, the envelope theorem says that we can compute the derivative of the optimal value function as

$$\left(\frac{\partial \pi}{\partial p}\right)_{(u,v)=(u^*,v^*)} = v^* = \frac{1}{15}(-700 + 800p) = \frac{-140 + 160p}{3}.$$

Q2. Show that a monomial of degree 6, $f(u_1, u_2, u_3) = u_1^2 u_2^3 u_3$ is a homogenous function of degree 6.

Ans.

$$f(tu_1, tu_2, tu_3) = (tu_1)^2 (tu_2)^3 (tu_3) = t^2 u_1^2 t^3 u_2 t u_3 = t^6 u_1^2 u_2^3 u_3 = t^6 f(tu_1, tu_2, tu_3)$$

The function $f(u_1, u_2) = \sqrt{u_1^3 + u_2^3}$ is a homogeneous function of degree $\frac{3}{2}$.

$$\mathrm{f}(\mathrm{u}_1, \mathrm{u}_2) = \sqrt{(\mathrm{tu}_1)^3 + (\mathrm{tu}_2)^3} = \sqrt{\mathrm{t}^3(\mathrm{u}_1^3 + \mathrm{u}_2^3)} = \mathrm{t}^{\frac{3}{2}}\sqrt{\mathrm{u}_1^3 + \mathrm{u}_2^3}.$$

Q3. Minimise $2u_1^2 + u_2^2$ Subject to: $u_1 + u_2 = 1$

Ans. The Lagrangian is:

$$L(u_1, u_2, \lambda_1) = 2u_1^2 + u_2^2 + \lambda_1(1 - u_1 - u_2)$$

Solve for the following:

$$\frac{\partial L(u_1^*, u_2^*, \lambda_1^*)}{\partial u_1} = 4u_1^* - \lambda_1^* = 0$$

$$\frac{\partial L(u_1^*, u_2^*, \lambda_1^*)}{\partial u_2} = 2u_2^* - \lambda_1^* = 0\,2$$

$$\frac{\partial L(u_1^*, u_2^*, \lambda_1^*)}{\partial \lambda} = 1 - u_1^* - u_2^* = 0$$

Solving this system of equations yields $u_1^* = \frac{1}{3}, u_2^* = \frac{2}{3}, \lambda_1^* = \frac{4}{3}$.